



Reinventing Turnarounds in the Metaverse.



How a metaverse-enabled digital twin could help plant operators reduce cost and risk.

For industrial plant operators, planned maintenance, upgrades or renewals can be complex and costly. The process of taking refineries or other types of units offline through turnarounds, or shutdowns and outages, is critical to ongoing operations and can account for a large portion of the planned maintenance budget. The work is expansive because it can involve interactions with numerous processing, work management and scheduling systems across operations as well as tight coordination of thousands of contractors and staff across reliability, maintenance, engineering, project management, safety, supply chain and emergency response teams.

Even with significant planning throughout the five key phases of the event — scoping and planning, procurement, onboarding, execution and post-turnaround management — unexpected costs and schedule delays can be common. Unplanned additional downtime could cost millions per day, may lead to significant lost revenue for the company and could have potentially far-reaching supply impacts for consumers. According to an Energy Information Administration (EIA) report, a 29-day turnaround at a US refinery cost an estimated \$39 million, involved 1,800 contractors and had the potential to lose between \$1.2 million to \$3 million for each day the turnaround went beyond the scheduled outage.

Poor communication, inaccurate cost estimation, lack of coordination and other unexpected issues during execution can be an underlying cause of these issues during turnaround management. To avoid cost overruns, anticipate and navigate issues and realize other operational improvements, some leading operators have turned to industrial metaverse-enabled digital twins, an emerging technology that has the potential to transform large-scale maintenance events. Although this new digital world is far from fully mature, these 3D visualizations of the existing power or industrial plant are already helping to provide a more holistic view of assets and real-time visibility — leading to better planning and coordination, improved efficiency, lower costs and reduced risk during turnarounds.

How a metaverse-driven digital twin can help throughout your turnaround.

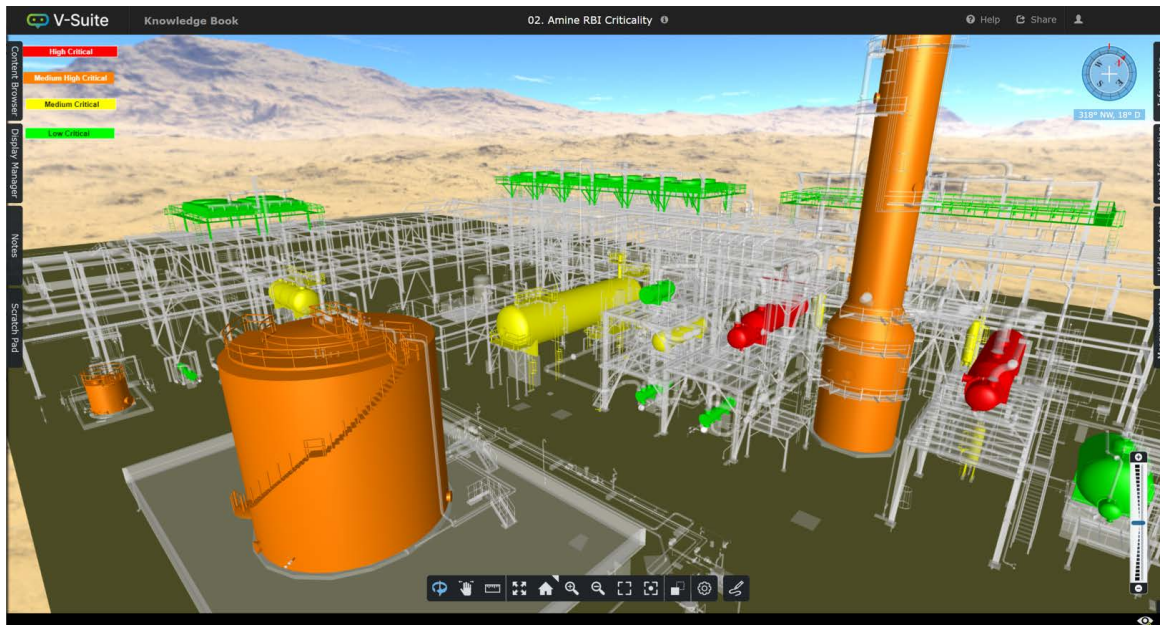
As the industrial metaverse develops, operational digital twins are emerging as a tool to help facilities execute turnarounds more efficiently. Digital twins are 3D visualizations of industrial plants, linked to piping and instrumentation diagrams, maintenance systems, process data, analytics software and data visualization tools. 3D digital twins can be built from computer-aided design (CAD) engineering and design models and updated with field scan data and field observations to stay relevant through the life of a plant. A metaverse-driven turnaround using a digital twin can transform the five key phases of a turnaround from the planning phase all the way to execution and post-turnaround management.

Scoping and planning: In the planning phase, engineers and planners could employ a condition-based maintenance approach by accessing critical data such as plant layouts, piping and instrumentation diagrams (P&ID), maintenance records, equipment dimensional details and other data from critical systems of record in real time directly from within the digital twin. This could accelerate and improve the planning process and optimizes work performed as part of the turnaround, helping to minimize downtime.

Figure 1 shows a refinery unit with data-driven colorization of assets. It has a risk criticality filter applied that is based on real-time equipment health data and maintenance records from key operational systems of record. During the planning phase of a turnaround, this technology enables engineers and maintenance teams to visualize the equipment layout and identify potential issues or bottlenecks that may arise during the turnaround. By analyzing the data from the digital twin, the operator can also identify areas where efficiencies can be improved and maintenance tasks can be optimized. Digital twin technology provides access to important information necessary to efficiently plan critical turnaround

functions, such as P&IDs, maintenance system and process data, analytics software and other data visualization tools. By utilizing digital twin technology, engineers and maintenance teams can streamline the planning phase of the turnaround, thereby reducing costs and minimizing downtime.

Figure 1: Display based on asset risk



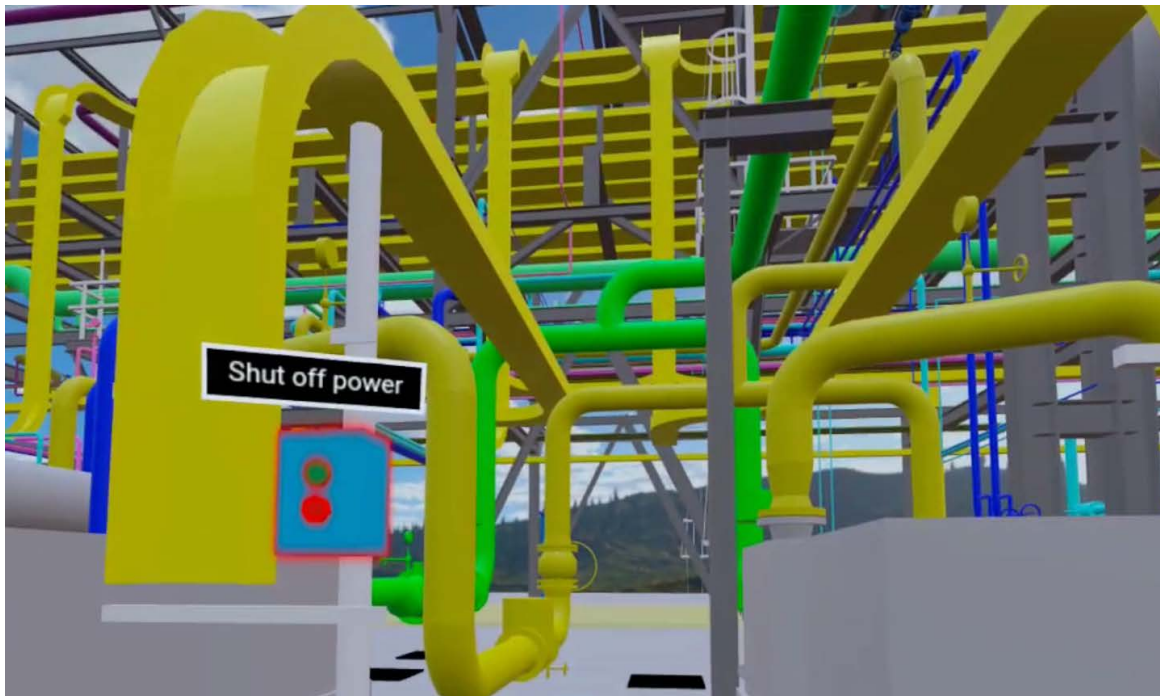
Procurement tracking: Once the plan is in place, the detailed digital twin can quickly identify the equipment, parts and materials required to execute the maintenance identified. Digital twins can help with the procurement of equipment and parts by linking inventory and maintenance plans, highlighting what's needed and allowing for easier orders through the connected system. Orders can then be tracked through the digital twin and any delays and/or changes are immediately highlighted. Remediation actions can be taken, as required, and plans updated easily to enable high visibility and a high degree of preparedness prior to execution.

Pre-turnaround onboarding: The metaverse-based digital twin can be used for onboarding and training of turnaround teams. Virtual reality

(VR), already well-known for offering realistic simulations for equipment maintenance, can be used more broadly to train contractors on plant layout as well as safety and maintenance procedures — all through a 3D visualization of the actual plant. The digital twin can also enable contractors and operators to practice procedures in virtual reality so they are familiar with the actual equipment they will be working on during the turnaround. The training will also list key tools and equipment needed for each turnaround task and help make sure vendors are well equipped and prepared to complete their scope of work.

Figure 2 shows an example of a virtual reality walk-through guide, where users can complete virtual walk-throughs in the digital twin. Contractors and operators can complete work procedures in the digital environment before going in the field, improving the effectiveness and speed of training while also minimizing burden on ongoing operations.

Figure 2: Virtual reality walk-throughs



Real-time execution monitoring: During this phase, the project leadership can monitor all the activities and personnel in real time using the digital twin to track the progress of the turnaround and highlight any areas that require immediate attention. Any delays against schedule are identified and immediately highlighted and escalated to ensure prompt remediation and timely completion.

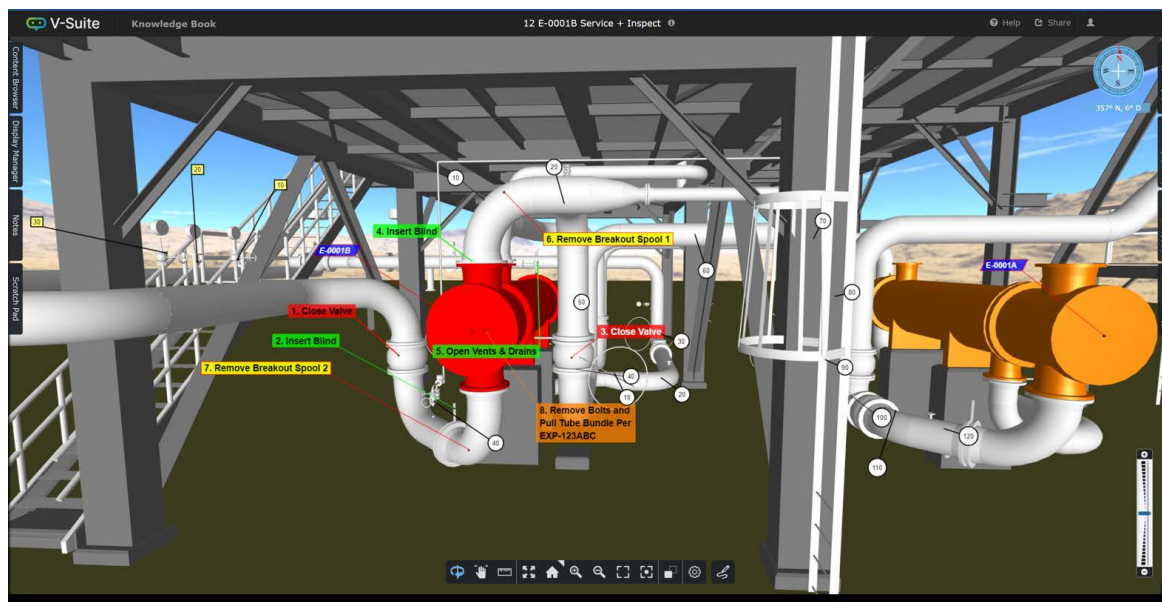
Safety and security are enhanced by tracking personnel and identifying any unauthorized activities. The model uses Internet of Things (IoT) and tracking sensors (e.g., RFID chips) with data integrated into the digital twin to continuously track all the activities and personnel. This allows turnaround leadership to monitor execution activities in real time. As contractors are accessing the plant, geofencing can be used to ensure they don't enter unauthorized or hazardous areas without proper authorization.

A virtual geospatial log of where contractors were and when during the turnaround helps to process accurate and faster time- and materials-based invoices based on the work performed. The geospatial log acts as a source of truth regarding how long activities took to perform and how many workers were needed for each task.

Post-turnaround management: Lastly, in post-turnaround management the digital twin can act as a single source of truth, providing detailed documentation for everything learned during the turnaround. It can enhance the accuracy and completeness of the documentation by facilitating real-time updates as work is being performed, ensuring that necessary information is captured and reducing the risk of errors or omissions. Additionally, pictures and videos of equipment can be tagged to the specific assets for future teams to reference, Light Detection and Ranging (LiDAR) scans of equipment can be uploaded to the digital twin, updated dimensional data and other observations can be logged in the model, tagged to the specific location in the plant. This ability to tag information, observations and procedures to specific locations in the 3D digital twin, as related to the physical locations in the plant, provides a more granular level of detail that can be useful for future maintenance activities.

Figure 3 shows an example of a work procedure saved to the digital twin, with the specific steps tagged to the related parts. Procedures developed during the turnaround can be saved this way and retained for use in future turnarounds or maintenance jobs.

Figure 3: Work procedure knowledge capture

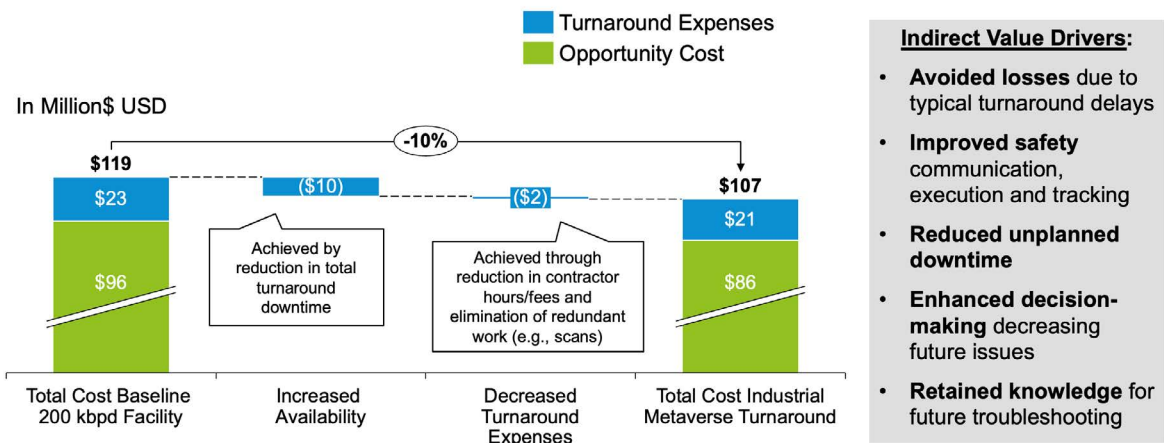


How the metaverse can drive value for plant operators.

Metaverse-driven turnarounds can be adjusted to focus on the activities needing attention in that cycle, facilitating faster and lower-cost turnarounds than traditional ones. This helps create value across three levels: reducing maintenance spend, increasing the overall availability of the facility and reducing future unplanned shutdowns. According to our estimates, a 200kbpd facility executing a 29-day turnaround may spend over \$20 million and lose up to \$100 million in opportunity cost at a \$17.50 crack spread, or refining margin. Leveraging the industrial metaverse can reduce turnaround costs by 5% to 10% and reduce turnaround time and therefore opportunity costs by 8% to 12%. These savings for a 200kbpd

(or 200,000 barrels per day) facility translate into \$2 million in turnaround expenses and \$10 million in increased revenue per turnaround; totaling \$12 million in value per turnaround. In addition to direct and immediate savings and increased availability to generate revenue, the industrial metaverse can help to reduce unplanned downtime, improve safety, enhance decision making and retain knowledge for future troubleshooting. These indirect levers can help drive additional value but aren't quantified in this analysis.

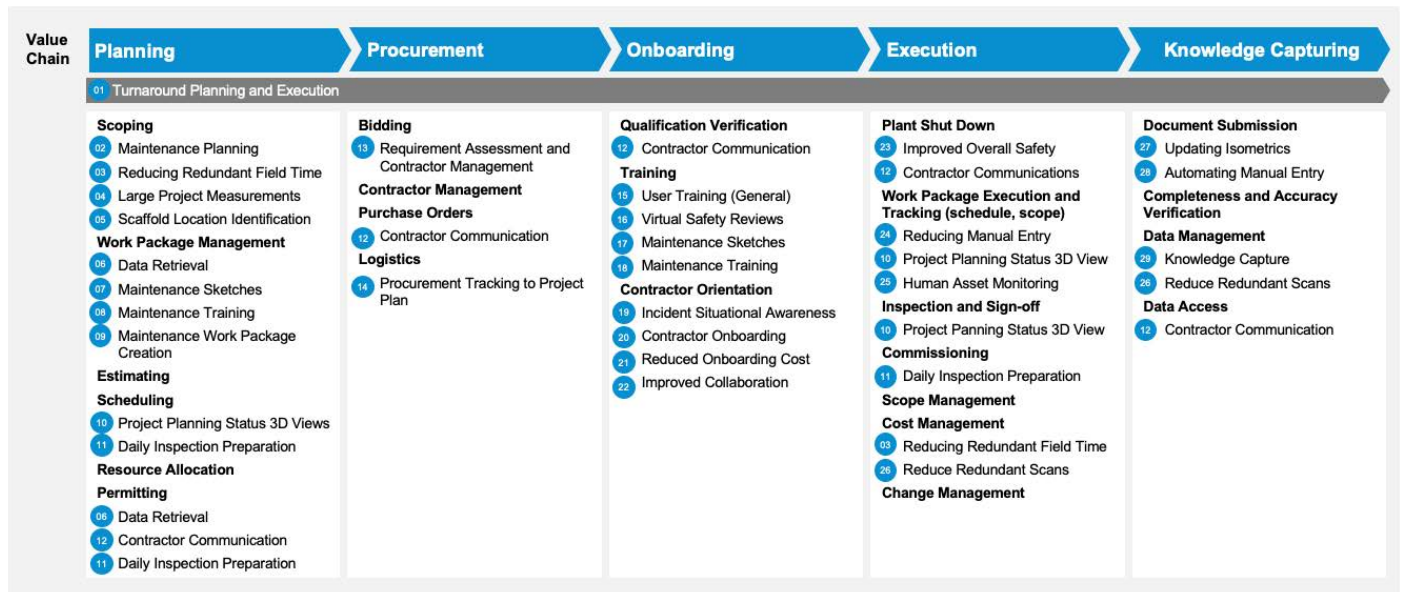
Figure 4: Industrial metaverse value pools



Key assumption: Cost savings for a 200kbp facility based on linear scaling from a 340kbp refinery case study with 29-day turnaround time

These savings are achieved by incorporating digital twin technology throughout the life cycle of a turnaround and fundamentally changing how turnaround teams operate. Redesigning turnaround work processes and leveraging the industrial metaverse may unlock multiple value drivers across a typical turnaround.

Figure 5: Industrial metaverse value drivers



Prioritize and expedite planning.

During the planning phase of the turnaround, engineering, maintenance and turnaround management teams go through scoping, work package management, estimating, scheduling, resource allocation and permitting for turnarounds. At this stage, the industrial metaverse helps with rightsizing and prioritizing turnaround activities. This could include assessing the equipment's health and performing a risk/failure analysis to determine which equipment needs immediate attention and which can be deferred to the next cycle. Specific value drivers include:

- **Reducing maintenance time:** Utilizing the digital twin as a central data repository and visual tool allows for quicker work identification, prioritization and tactical execution planning. This can save an estimated 1,000 hours or more of maintenance hours throughout the pre-turnaround planning phase.
- **Accelerating equipment planning:** The digital twin helps turnaround planners quickly make decisions on location planning, like where cranes, equipment or scaffolding should be located. Looking at a plant through a 3D lens helps planners identify risks or hazards that could

arise from the placement of large equipment/structures and reduces the amount of rework that happens during the execution phase — something that would otherwise add cost and delay the overall turnaround timeline.

- **Producing usable, organized data:** Engineering records (e.g., drawings), traditionally saved on company-shared drives are critical to effective and efficient planning. According to our estimates, maintenance and engineering teams can spend up to 50% of their time trying to locate accurate data. In many cases, the most up-to-date records cannot be located in a timely manner due to insufficient retention and governance procedures. In the industrial metaverse, data elements are intuitively organized by assets. When navigating to the desired equipment, the most up-to-date engineering records can be located with one click within the model. This helps reduce the time spent locating engineering records and enables up-to-date information is available to engineers for effective decision-making.

The industrial metaverse has additional value cases throughout the planning phase, enabling each set of activities to be done faster, with fewer people and completed more effectively than traditional turnaround planning practices.

Source and secure resources through enhanced procurement.

Procurement is a critical component of pre-turnaround activity including bidding, contractor management, purchase orders and logistics and must be managed well to help make sure an on-schedule and under-budget turnaround. Equipment, material and contractor availability are critical to turnaround execution and timeliness. Operating companies must be able to make timely procurement decisions and secure resources ahead of time for large-scale and complex turnarounds to reduce execution risk and save cost. Metaverse-driven turnarounds create value at this stage through:

- Improving requirements assessment and contractor management: Work packages and turnaround complexity can easily be communicated to contractors through the metaverse, resulting in an early kickoff of the formal procurement process (e.g., request for proposals, bidding, contract finalization). Contracts can be written using the virtual 360-degree functionality of the digital twin, reducing overall field visits to the plant and enabling more contractors to bid on work items. This creates value for operating companies by allowing sufficient time to compare options, negotiate prices and locate long-lead-time equipment and services. In some cases, it also offsets the seasonality of turnarounds where multiple large-scale turnarounds are being planned in the region and material and services become scarce.
- Enhancing procurement tracking: The digital twin tags procurement and inventory status to assets allowing engineers and maintenance teams to easily identify what needs to be ordered and the status of each part. Centralizing procurement tracking in a single planning tool helps teams stay up to date on the status of critical path pieces of equipment and address any delays or impacts in real time as they occur, minimizing any unforeseen impacts to the overall turnaround schedule.

The industrial metaverse integrates contractor communication and procurement tracking into the everyday planning work processes of turnaround teams, leading to more effective internal and external collaboration and improved value of contractors/equipment.

Virtually transform onboarding and training •

The onboarding stage of a turnaround includes qualification verification, training and contractor onboarding. All three of these activities are transformed in a metaverse-driven turnaround by leveraging 3D visualizations of the plant and virtual reality technology. Specific value cases include:

- **Accelerating contractor onboarding:** Leverage virtual orientation and training to get contractors up to speed faster on the work they will perform. Using digital tools to communicate and train contractors on work packages can decrease the time they are needed in the field, creating value by reducing the amount of time contractors are needed and decreasing distractions for operators pre-turnaround. Effective onboarding and reduced field time can significantly improve the safety record for turnarounds.
- **Decreasing time spent on maintenance training:** Enable new trainees to take a virtual orientation of the facility and get introduced to the various assets, processes and facilities involved in a turnaround. Maintenance staff can be trained on procedures within the metaverse and walk through specific tasks they will perform in the field, completing them virtually first. This creates value by decreasing the time it takes to train maintenance staff and improving the effectiveness of their work — thereby decreasing any issues in the field that could cause delays in the turnaround.
- **Enhancing safety reviews:** Use virtual reality tools to enable contractors and maintenance teams to navigate through the digital twin and review locations of shelter in place facilities, safety showers, emergency shut-off locations, hazardous equipment locations, hazardous shock zones, toxic material zones and any other site-specific safety reviews. This creates value by increasing the effectiveness of safety reviews thereby lowering risk, decreasing the time spent on training and reducing the amount of time contractors are needed on-site by allowing contractors and maintenance teams

to familiarize themselves with the site-specific safety protocols well before the actual turnaround.

- **Reducing onboarding costs:** By leveraging digital tools to communicate with and train contractors, the need for in-person training sessions and materials can be reduced, resulting in cost savings for the organization. Additionally, virtual training can help reduce the amount of time needed to onboard new contractors, freeing up resources for other critical activities.
- **Improving collaboration:** The use of virtual models can facilitate improved collaboration between contractors and the facility's staff. Virtual collaboration spaces can be set up to allow for real-time discussions, remote walk-throughs and joint problem-solving sessions, resulting in a more cohesive and efficient team. Improved collaboration can lead to better outcomes and reduce the likelihood of delays during turnarounds.

Traditionally, pre-turnaround onboarding and orientation activities are limited to off-site document reviews and generic and sometimes outdated online training (e.g., safety regulations). Actual job orientation is conducted after the turnaround starts. In the industrial metaverse virtual reality enables an interactive experience for operators, contractors and service providers to practice maintenance procedures, get tools ready and familiarize themselves with safety protocols well before the actual turnaround. This can change how teams prepare for turnarounds and the effectiveness of everyone involved.

Improve execution monitoring and tracking.

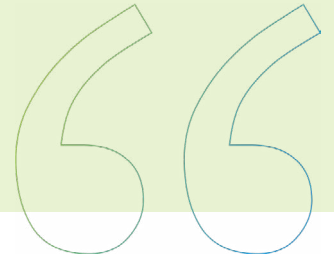
The execution phase consists of seven key components: plant shutdown, work package execution and tracking, inspection and sign-off, commissioning, scope management, cost management and change management. Large-scale turnaround execution is complex with hundreds

of work packages, unanticipated events/interruptions and thousands of contractors on site. The industrial metaverse helps manage and reduce complexity through a series of value cases:

- **Accelerating turnarounds through real-time tracking:** Turnaround progress tracking has historically been manually reported by teams and tracked by project management tools requiring manual data entry, tracking and updates. Timeliness and detailed accuracy often lag. The industrial metaverse enables automated tracking, visualization of work progress in virtual reality and management of work traffic on site. Digital twins can color-code assets by complete, at risk/potentially delayed, or delayed with root causes captured. Improved tracking and visualizations of turnarounds will create value by improving coordination and identification of issues so that they can be resolved without impact to timeline.
- **Reducing manual entry:** Centralization and visualization of asset data, including isometrics and internal dimensions, can reduce the need for manual entries during a turnaround. This generates value by reducing the time and resources required, avoiding congestion, lowering safety risk and leading to a shorter turnaround time and reduced downtime for the facility.
- **Improving overall safety:** Teams can leverage the industrial metaverse to complete daily safety reviews, discuss real-time hazards and learn while referencing specific locations in the digital twin. This creates value by improving communication and understanding of safety hazards, streamlining daily safety reviews, improving overall safety and allowing workers to spend more time completing critical path work for the turnaround.

Execution of turnarounds can happen more efficiently as the industrial metaverse improves planning in previous stages and streamlines execution of activities. The digital twin reduces the need for certain data-gathering steps and improves the effectiveness of communication among teams.

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Capture knowledge and process improvements.

In a turnaround, many modifications of the plant happen during a short time frame. Therefore, post-turnaround knowledge capturing is critical for companies to maintain the knowledge base for root-cause analysis, future lessons learned, turnaround process improvement and maintaining the most up-to-date engineering records. There are four key steps for knowledge capturing: document submission, completeness and accuracy verification, data management and ongoing data access. Traditionally, the first two steps are manually completed, which can put an added burden on engineers, operators and maintenance personnel when they should be focused on fast-paced executions and safety. During extremely busy times, these steps can be overlooked or skipped, which makes post-turnaround knowledge capturing extremely challenging and introduces long-term and significant compliance and liability risks to the operating company. The industrial metaverse streamlines the knowledge-capture process, creating a central repository and single source of truth, thus creating value in several ways:

- **Retention of isometrics and data:** The digital twin acts as a live repository of all isometrics, allowing for easy upload and long-term retention of new data. Streamlining this process creates value by reducing the time it takes engineers and maintenance teams to save this information and to look for it later when they are troubleshooting an issue.
- **Improved knowledge capture:** Turnarounds are a great time to take pictures and videos of opened or offline equipment to learn how they are designed. This information can be tagged to specific assets in the digital twin and retained for future use. Workers can also tag best or worst practices to assets or procedures to improve maintenance practices and allow for continued learning across turnarounds. The digital twin centralizes and stores this information in an easy-to-use location.
- **Reduction of redundant scans:** Contractor scans that are taken during a turnaround can be uploaded to the industrial metaverse to update equipment dimensions and store results for future reference. This information is frequently lost, so storing it in a centralized tool creates value by reducing the need for future scans, saving future turnarounds time and costs.

The industrial metaverse transforms the knowledge capturing stage of a turnaround by streamlining data submission/storage and creating a new way for teams to interact with existing data. This allows teams to get more out of their data and avoid re-collecting data that already exists within the digital twin.

With more than 25 value cases across the life cycle of a turnaround, the industrial metaverse truly transforms how work is completed and how turnarounds are conducted. Companies that successfully integrate these value cases into their work processes are reducing the overall cost of a turnaround by 10% and will drive additional value as the functionality and adoption of the industrial metaverse expands and additional use cases are developed.

Case study.

In an ARC Advisory Group study, a US refinery was able to achieve 8% to 10% faster return-to-production through better planning and execution. Early identification of projected work density conflicts, movement of temporary assets (e.g., cranes), more efficient facilitation of work tasks, improved construction sequencing, execution monitoring and prefabrication were cited as benefits of leveraging a metaverse-driven turnaround. This diesel and gasoline refinery leveraged many of the detailed value drivers listed above, from faster data retrieval for isometrics and measurements to maintenance training and knowledge capture. This facility continues to develop more use cases as it drives increased adoption and incorporates new information. Based on the four-day faster turnaround, opportunity cost recapture was estimated at \$7.5 million.

Key takeaways.

Many organizations are beginning to experiment with digital twins as a tool to improve productivity and efficiency of a turnaround. As the metaverse use grows, it can be important to continue evaluating applications throughout the organization and other potential impacts. For instance, while the metaverse can significantly reduce risk, it may also introduce new risk considerations that should be evaluated and managed through holistic risk management. It can be important to bring together operations, cybersecurity, information security and other leaders to establish guiding principles, governance, controls and other measures necessary to facilitate a secure and responsible use of technology.

Integration of these immersive technologies into standard work processes can start the journey into the industrial metaverse. Those who pursue using a metaverse-enabled digital twin for complex maintenance may realize labor and productivity benefits across the distinct phases of a turnaround addressed in this paper and summarized here in Figure 6.

Figure 6: Summary of industrial metaverse possible benefits

	Summary	Labor benefits	Productivity benefits
Planning	Improved project planning and coordination can reduce the need for manual data gathering and entry, saving time and reducing errors.	Reduction in labor hours for project management tasks	Improvement in planning accuracy, leading to fewer schedule delays and reduced costs
Procurement	Centralized and digital procurement processes can streamline approvals and improve coordination with vendors, reducing delays and saving time.	Reduction in labor hours for procurement tasks	Reduction in procurement lead times and improved viability leading to fewer delays and reduced costs
Onboarding	Reduced manual data entry, streamlined training and improved communication between teams may help to reduce errors and improve safety.	Reduction in labor hours for onboarding tasks	Reduction in training time and improved training effectiveness, leading to faster ramp-up times and improved productivity
Execution	Improved coordination, enabled real-time tracking, reduced manual data entry and enhanced communication can lead to faster execution and improved safety.	Reduction in labor hours for execution tasks	Reduction in turnaround time, leading to reduced downtime and increased productivity
Knowledge Capture	Streamlined data submission and storage, reduced redundant scans and centralized data can enable easier access and improved maintenance practices.	Reduction in labor hours for knowledge capture tasks	Improvement in ongoing maintenance practices, leading to reduced downtime and increased productivity

Leveraging industrial metaverse beyond turnarounds.

The application of the industrial metaverse in turnarounds is clear, and companies are already utilizing the technology to realize value. Other use cases exist across nearly all aspects of operations. For instance, integrating a digital twin into work processes can also help drive value in the following areas:

- Unplanned downtime
- Day-to-day troubleshooting
- Operations optimization
- New plant design
- Supply chain optimization
- Advanced process control
- Decommissioning
- Asset sale

Future use cases of the industrial metaverse span the entire asset life cycle. Operators can swiftly capture value pertaining to turnarounds by integrating systems and data into digital twin technology and redesigning work processes to help transform how work is done. Once the digital twin is set up and the benefits of the industrial metaverse realized, organizations can expand adoption across operations including response to unplanned outages, day-to-day troubleshooting, operational improvements and new plant design. The industrial metaverse is in the infancy of adoption, but already has the potential to drive change and process improvements across the industry.

Want to start capturing value with the industrial metaverse?

Connect with an expert below to find out how.

This report was a collaboration between PwC and Visionaize.

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Reference: Energy Information Administration Office of Oil and Gas U.S. Department of Energy, Refinery Outages: Description and Potential Impact on Petroleum Product Prices, March 2007

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